

CURRICULUM VITAE

KELLY A. DYER

ADDRESS:

Department of Genetics
University of Georgia
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EDUCATION:

- 2004 Ph.D. Evolutionary Genetics, University of Rochester, Rochester, New York
Advisor: John Jaenike
Thesis: Population genetics of sex ratio distortion in *Drosophila*
2001 M.S. Ecology and Evolutionary Biology, University of Arizona, Tucson, Arizona
1996 B.S. Evolution and Ecology (Highest Honors), University of California, Davis
Advisors: Michael Turelli and Donald Strong

CURRENT POSITION:

11/2007- Assistant Professor
Department of Genetics, University of Georgia

PREVIOUS ACADEMIC POSITIONS:

- 2005-2007 Royal Society USA Research Fellow
Institute of Evolutionary Biology, University of Edinburgh
Advisor: Brian Charlesworth
2004-2005 Postdoctoral Research Fellow
Department of Biology, University of Rochester
Advisor: John Jaenike
1996-1998 Senior Research Technician
Department of Ecology and Evolution, University of Chicago
Advisor: Jerry Coyne
1996 Post-Graduate Researcher
Center for Population Biology, University of California at Davis
Advisor: Michael Turelli

RESEARCH INTERESTS:

Population genetics and molecular evolution
Genetics of adaptation and speciation
Host-parasite interactions
Insect ecology and evolution

HONORS:

2009-2013	New Scholar in Aging, Ellison Medical Foundation
2009-2011	Lilly Teaching Fellow, University of Georgia
2009-2010	Powe Junior Faculty Enhancement Award, Oak Ridge Associated Universities
2009	OVPR Research Fellow, University of Georgia
2005-2007	USA Research Fellowship, The Royal Society
2005	Walter Fitch Award Symposium, Molecular Evolution (SMBE) meeting
2003	Edward Peck Curtis Award for Excellence in Teaching, University of Rochester (One of five university-wide annual recipients)
2003-2004	Dissertation Fellowship, American Association of University Women
2003	Workshop on Molecular Evolution Partial Scholarship, Woods Hole, MA
2001	Theodore Roosevelt Memorial Fund Grant, American Museum of Natural History
1999-2002	Graduate Research Fellowship, National Science Foundation
1998-2003	Caspari Fellowship, University of Rochester
1996	Departmental Citation for Excellence in Evolution and Ecology, UC Davis
1996	Phi Sigma National Honors Society, Delta Gamma Chapter, UC Davis

PUBLICATIONS:

19. **Dyer, K. A.** 2011. Local selection underlies the geographic distribution of *sex-ratio* drive in *Drosophila neotestacea*. *Evolution*, in press.
18. **Dyer, K. A.**, C. Burke, and J. Jaenike. 2011. *Wolbachia*-mediated persistence of mtDNA from a potentially extinct species. *Molecular Ecology* 20: 2805-2817.
17. **Dyer, K. A.**, B. E. White, M. J. Bray, D. G. Piqué, and A. J. Betancourt. 2011. Molecular evolution of a Y to autosome translocation. *Molecular Biology and Evolution* 28: 1293-1306.
16. Jaenike, J. and **K. A. Dyer**. 2008. No resistance to male-killing *Wolbachia* after thousands of years of infection. *Journal of Evolutionary Biology* 21: 1570-1577.
15. Unckless, R. L., L. M. Boilio, M. Schifano, and **K. A. Dyer**. 2008. Isolation and characterization of 30 polymorphic microsatellites from the mycophagous fly *Drosophila innubila*. *Molecular Ecology Resources* 8: 939-942.
14. **Dyer, K. A.** 2007. Identification and characterization of 21 polymorphic microsatellite loci from the mycophagous fly *Drosophila neotestacea*. *Molecular Ecology Notes* 7: 1120 – 1122.
13. **Dyer, K. A.**, B. Charlesworth, and J. Jaenike. 2007. Chromosome-wide linkage disequilibrium as a consequence of meiotic drive. *Proceedings of the National Academy of Science USA* 104: 1587-1592.
12. Jaenike, J., **K. A. Dyer**, C. Cornish*, and M. S. Minhas. 2006. Asymmetrical reinforcement and *Wolbachia* infection in *Drosophila*. *PLoS Biology* 4: 1852-1862.

11. **Dyer, K. A.** and J. Jaenike. 2005. Evolutionary dynamics of a spatially structured host-parasite association: *Drosophila innubila* and male-killing *Wolbachia*. *Evolution* 59: 1518-1528. (Selected by Faculty of 1000)
10. **Dyer, K. A.**, M. S. Minhas, and J. Jaenike. 2005. Expression and modulation of embryonic male-killing in *Drosophila innubila*: Opportunities for multi-level selection. *Evolution* 59: 838-848.
9. Shoemaker[^], D. D., **K. A. Dyer**[^], M. Ahrens, K. McAbee, and J. Jaenike. 2004. Decreased diversity but increased substitution rate in host mtDNA as a consequence of *Wolbachia* endosymbiont infection. *Genetics* 168: 2049-2058. ([^]contributed equally)
8. **Dyer, K. A.** and J. Jaenike. 2004. Evolutionary stable infection by a male-killing endosymbiont in *Drosophila innubila*: Molecular evidence from the host and parasite genomes. *Genetics* 168: 1443-1455. (Selected by Faculty of 1000)
7. Guindon, S., A. G. Rodrigo, **K. A. Dyer**, and J. P. Huelsenbeck. 2004. Modeling the site-specific variation of selection patterns along lineages. *Proceedings of the National Academy of Science USA* 35: 12957-12962.
6. Huelsenbeck, J. P. and **K. A. Dyer**. 2004. Bayesian estimation of positively selected sites. *Journal of Molecular Evolution* 58: 661-672.
5. Jaenike, J., **K. A. Dyer**, and L. K. Reed. 2003. Within-population structure of competition and the dynamics of male-killing *Wolbachia*. *Evolutionary Ecology Research* 5: 1023-1036.
4. Ross, C. L., **K. A. Dyer**, T. Erez, S. J. Miller, J. Jaenike, and T. A. Markow. 2003. Rapid divergence of microsatellite abundance among species of *Drosophila*. *Molecular Biology and Evolution* 20: 1143-1157.
3. Price, C. S. C., **K. A. Dyer**, and J. A. Coyne. 1999. Sperm competition in *Drosophila* males involves both displacement and incapacitation. *Nature* 400: 449-452.
(Featured on journal cover and highlighted by a 'News and Views' article by T. Birkhead)
2. **Dyer, K.** 1998. Darwin on Disk (A review of the Second Edition of the Darwin CD-ROM). *Evolution* 52: 295.
1. Strong, D. R., H. K. Kaya, A. V. Whipple, S. Kraig, M. Bondonno, **K. Dyer**, and J. Maron. 1996. Entomopathogenic nematodes: Natural enemies of root-feeding caterpillars on bush lupine. *Oecologia*, 108: 167-173.

COURSES TAUGHT AT UGA:

Spring 2011 GENE 8160. Advanced Evolutionary Genetics
Spring 2011 GENE 4230L. Evolutionary Biology Laboratory
Spring 2010 GENE 8160. Advanced Evolutionary Genetics
Fall 2009 GENE 4230L. Evolutionary Biology Laboratory
Spring 2009 GENE 8950. Molecular Evolution

Fall 2008 GENE 8700. Current Topics in Evolutionary Genetics
Spring 2008 GENE 4950. Senior Seminar in Genetics

PROFESSIONAL SERVICE:

2012 Co-Organizer, Evolution and Quantitative Genetics Platform Session,
52nd Annual *Drosophila* Research Conference
2011-2013 Elected Council Member, American Genetics Association
2011 Grant Panel Member, National Science Foundation
2010-2012 Associate Editor, *Evolution*
2009-2012 Reviewing Editor, *Journal of Evolutionary Biology*
2008 Co-organizer, SouthEastern Population Ecology and Evolutionary Genetics
(SEPEEG) meeting. 100 attendees, held in Eatonton GA.
2005 Selection committee member, Workshop on Molecular Evolution, Wood's Hole
2005-current Invited peer reviewer: *American Naturalist*, *Behavioral Ecology and*
Sociobiology, *Behavior Genetics*, *BMC Evolutionary Biology*, *Ecological*
Research, *Evolution*, *Genetics*, *Genetical Research*, *Genome Research*,
Heredity, *Journal of Theoretical Biology*, *Journal of Evolutionary Biology*,
PLoS Genetics, *PLoS One*, *Proceedings of the Royal Society Series B*,
Science, Natural Environment Research Council (UK), Biotechnology and
Biological Sciences Research Council (UK), French National Research
Agency

SYNERGISTIC ACTIVITIES:

2012 Invited participant, "Tracking the biotic response to global climate change through
genomic analysis" Nescent Catalysis group
2009 Invited participant, "Intragenomic conflict," NSF National Institute of Mathematical
and Biological Synthesis (NIMBioS) working group

SELECTED UNIVERSITY AND DEPARTMENTAL SERVICE:

2010-current Graduate advisory committee, Department of Genetics
2010-2011 Executive committee, Department of Genetics
2010-2011 CURO Expansion faculty committee, Honors College
2010 Genetics department head search committee, Franklin College and Genetics
2009-current Curriculum committee, Department of Genetics
2009-current James L. Carmon Scholarship selection committee, OVPR
2008, 2011 Genetics graduate recruitment planning committee, Department of Genetics
2008-2010 Franklin College Faculty Senate, Genetics representative
2009-2010 Chair, Academic Standards committee, Franklin College Faculty Senate
2008-2010 Admissions Committee, Franklin College Faculty Senate
2008-current Co-organizer and co-founder, Enthusiasts of Diversity, Genetics, and Evolution
(EDGE), a weekly, inter-departmental informal seminar series
2008 Goldwater nomination committee, Honors College
2008 Plant population genetics search committee

STUDENT SUPERVISION AND COMMITTEES WHILE AT UGA:

Primary advisor for two PhD students in Genetics
First year advisor for three PhD students in Genetics
Graduate rotation research advisor for five PhD students (four Genetics, one ILS)
Member of 15 PhD committees at UGA (Genetics and Plant Biology) and Emory University
External evaluator for one PhD thesis (University of Melbourne)
Research mentor for 21 undergraduate researchers:

- advisor for five CURO Honors Theses and two CURO summer fellows
- nine URM students and three LSAMP research scholars
- two students in summer REUs and one in the Young Dawgs program

Member of two undergraduate CURO Honors thesis committees

SELECTED OUTREACH:

2009-2011	Judge, Georgia Science and Engineering Fair
2009, 2010	Host lab for HONS 3070H class – Introduction to Research
2011	EDEN (Eco-Devo-Evo Network) selection committee member for undergraduate summer fellowships
2009	Judge for poster awards, Molecular Evolution (SMBE) meeting
2008	Undergraduate diversity program mentor, Molecular Evolution (SMBE) meeting
2008	“Careers in STEM” panelist, UGA Career Center symposium

COURSE PARTICIPANT:

2010	Experimental Aging Research Summer Course, Barshop Institute, University of Texas Health Sciences Center, San Antonio
2008	Summer Institute in Statistical Genetics, University of Washington QTL mapping and Advanced QTL mapping modules
2006	Science Communication Training Course, Royal Society, London, UK
2003	Workshop on Molecular Evolution, Woods Hole MBL
1996	University of California Education Abroad in Tropical Biology in Costa Rica
1995	Bodega Marine Laboratory, UC Davis field ecology course and research